

Ex. No.: 7

Date: 26/09/2025

IPC USING SHARED MEMORY

Aim:

To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process.

Algorithm:

sender

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Write a string to the shared memory segment using sprintf
5. Set delay using sleep
6. Detach shared memory segment using shmdt

receiver

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Print the shared memory contents sent by the sender process.
5. Detach shared memory segment using shmdt

Program Code:

sender.c

```
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#define SharedMemSize 50
```



```

void main() {
    char c;
    int shmid;
    key_t key;
    char * shared-memory;
    key = 5677;
    if ((shmid = shmget (key, shared mem size, IPC_CREAT |
        0666)) < 0)
    {
        perror ("shmget");
        exit(1);
    }
    if ((shared-memory = shmat (shmid, NULL, 0)) ==
        (char *) -1)

        perror ("shmat");
        exit (1);
    printf (shared-memory, "Welcome to Shared Memory");
    sleep(2);
    exit(0);
}

```



```

receiver
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>
#define SHARED_MEM_SIZE 50

void main() {
    int shmid;
    key_t key;
    char * shared-memory;
    key = 5677;
    if ((shmid = shmget(key, SHARED_MEM_SIZE, 0666)) < 0) {
        perror("shmget");
        exit(1);
    }

    if ((shared-memory = shmat(shmid, NULL, 0)) == (char *)-1) {
        perror("shmat");
        exit(1);
    }

    printf("Message Received %s\n", shared-memory);
    exit(0);
}

```


Sample Output

Terminal 1

```
[root@localhost student]# gcc sender.c -o sender
[root@localhost student]# ./sender
```

Terminal 2

```
[root@localhost student]# gcc receiver.c -o receiver
[root@localhost student]# ./receiver
Message Received: Welcome to Shared Memory
[root@localhost student]#
```

gcc sender.c -o sender
gcc receiver.c -o receiver

o/ sender

sender: Data written to shared memory
welcome to shared memory

o/ receiver:

receiver: Data read from shared memory
Welcome to sender process.

Result:

Thus IPC using shared memory between sender process and receiver process has been executed successfully